

Installing Qulacs on Windows (Anaconda)

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Step 0: Open Anaconda Prompt

- ▶ All commands must be run in **Anaconda Prompt**, not **regular CMD**.
- ▶ This ensures proper environment activation and package management.

Step 1: Create a new Conda environment

```
conda create -n qulacs_env python=3.10 -y
```

- ▶ Creates a new environment called `qulacs_env` with Python 3.10. 

Step 2: Activate the environment

```
conda activate qulacs_env
```

- ▶ Activate your new environment.



Step 3: Install required build tools

```
conda install -c conda-forge cmake wheel  
setuptools ninja -y
```

- ▶ **cmake** → for building extensions if needed
- ▶ **wheel & setuptools** → Python packaging tools
- ▶ **ninja** → speeds up C++ builds 

Step 4: Install Qulacs (pre-built wheel)

```
pip install qulacs --only-binary :all:
```

- ▶ Using `--only-binary :all:` ensures it **does not build from source**. 
- ▶ Avoids common CMake errors on Windows.

Step 5: Add Jupyter kernel

```
conda install ipykernel -y
python -m ipykernel install --user --name=
    qulacs_env --display-name "Python (Qulacs)"
```

- ▶ Adds qulacs_env to Jupyter Notebook as a selectable kernel.



Step 6: Test Qulacs

Open Jupyter Notebook, select kernel **Python (Qulacs)**, and run:

```
import qulacs
print(qulacs.__version__)
```

- ▶ You should see the Qulacs version printed. 
- ▶ If no errors occur, installation is successful!

All-in-One Command (One-liner)

```
conda create -n qulacs_env python=3.10 -y && ^
conda activate qulacs_env && ^
conda install -c conda-forge cmake wheel
    setuptools ninja ipykernel -y && ^
pip install qulacs --only-binary :all: && ^
python -m ipykernel install --user --name=
    qulacs_env --display-name "Python (Qulacs)"
    && ^
echo Qulacs installation complete! Run "jupyter
    notebook" and select kernel "Python (Qulacs)"
```

- Complete installation in one command.